

Integrals – 5

The gamma and beta functions and Wallis' formulas

1 сентября 2026 года

Theoretical minimum.

Def. $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$ for $\operatorname{Re} s > 0$, and $B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx$.

Theorem 1 (basic properties). $\Gamma(s+1) = s\Gamma(s)$, $\Gamma(n+1) = n!$, $B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$, the reflection formula $\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}$ and Legendre's duplication formula $\Gamma(s)\Gamma(s + \frac{1}{2}) = 2^{1-2s} \sqrt{\pi} \Gamma(2s)$.

Theorem 2 (Wallis). $\int_0^{\pi/2} \sin^n x \, dx = \int_0^{\pi/2} \cos^n x \, dx = \frac{\sqrt{\pi}}{2} \cdot \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2} + 1)}$,
whence $\frac{\pi}{2} = \prod_{n \geq 1} \frac{(2n)^2}{(2n-1)(2n+1)}$.

Fact 1. The substitution $x = \sin^2 t$ turns the beta integral into a trigonometric one, $B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} t \cos^{2q-1} t \, dt$, and $x \mapsto \frac{x}{1+x}$ turns it into $\int_0^\infty \frac{x^{p-1}}{(1+x)^{p+q}} dx$.

1. Prove that $B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$ (a double integral and a passage to polar coordinates) and deduce $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.
2. Derive both Wallis formulas from Theorem 2 and compute $\int_0^{2\pi} \cos^{2020} x \, dx$ (Integration Bee 2020) and $\int_0^\pi \sin^2 x \cos^2 x \, dx$ (Integration Bee 2013).
3. (Integration Bee) Beta integrals in disguise:
 - a) $\int_0^1 \sqrt{x(1-x)} \, dx$ (2025);
 - b) $\int_0^1 \sqrt{1-x^2} \, dx$ (2020);
 - c) $\int_0^1 \sqrt{1-\sqrt{x}} \, dx$ (2022);
 - d) $\int_0^1 x^{2024} (1-x^{2025})^{2025} dx$ (2025).

4. Compute $\int_0^\infty \frac{x^{a-1}}{1+x} dx = \frac{\pi}{\sin \pi a}$ for $0 < a < 1$ through the beta function and the reflection formula; compare with the contour computation in the sheet Complex Analysis – 4.
5. (Integration Bee 2019) Compute $\lim_{n \rightarrow \infty} \int_{-\infty}^\infty e^{-x^{2n}} dx$ and explain the answer geometrically; express the integral itself through Γ .